

Beta function:

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

two variables / power

8) $\int_0^1 x^9 (1-\sqrt{x})^5 dx$

$$u = \sqrt{x} \quad ; \quad x = u^2$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$dx = 2\sqrt{x} du$$

$$\int_0^1 (u^2)^9 (1-u)^5 \cdot 2\sqrt{x} du$$

$$\int_0^1 u^8 (1-u)^5 \cdot 2u du$$

$$2 \int_0^1 u^9 (1-u)^5 du$$

$$2 \int_0^1 u^{10-1} (1-u)^{6-1} du$$

$$2 \beta(10, 6) du$$

Relationship between Beta and Gamma function

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

$$2 \beta(10, 6) = \frac{\Gamma(10) \cdot \Gamma(6)}{\Gamma(16)}$$

$$= 2 \cdot \frac{9! \times 5!}{15!}$$

$$= \frac{1}{1505}$$

$$e) \int_0^1 x^2 (1-x^2)^3 dx$$

$$e) \int_0^1 \sqrt{x} (1-x)^{\frac{1}{2}} dx$$

$$e) \int_0^1 x^3 (1-x)^{\frac{2}{3}} dx \quad \int_0^1 n^{4-1} (1-n)^{\frac{7}{3}-1} dn$$

$$B(4, \frac{7}{3}) \Rightarrow \frac{\sqrt{4} \sqrt{7/3}}{\sqrt{19/3}} \Rightarrow$$

$$3! \cdot \frac{7}{3}$$

$$\frac{16}{3} \cdot \frac{13}{3} \cdot \frac{10}{3} \cdot \frac{7}{3} \sqrt{\frac{7}{3}}$$

$$\Rightarrow \frac{6}{16 \cdot 13 \cdot 10 \cdot 7} \\ 81$$

$$\sqrt{n+1} = n\sqrt{n}$$

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